

Automatic Image Analysis
Questions for exam preparation
Part II

62. Which properties need a function to fulfill to be a probability?
 63. What are the differences between a probability and a probability density?
 64. What is the joint probability of two events?
 65. What is the conditional probability of an event A given B?
 66. What is marginalization?
 67. How can the joint probability of two events be factorized into probabilities of the individual events?
 68. If two events are independent, what is then known about their conditional and joint probabilities?
 69. What is Bayes' Theorem?
 70. What are the technical names of all terms in Bayes' theorem?
 71. Given the joint probability of two events, derive Bayes' theorem.
 72. What does MAP stand for?
 73. What is the decision rule of a MAP estimator?
 74. What does ML stand for (in the context of probability theory)?
 75. What is the decision rule of a ML estimator?
 76. In which case are ML and MAP equivalent?
 77. What is Naive Bayes?
 78. How is it possible to take the cost of a decision into account?
 79. What is a discriminant function?
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80. What is the difference between parametric and nonparametric methods of probability estimation?
 81. What are examples for nonparametric approaches?
 82. What are examples for parametric approaches?
 83. What is the curse of dimensionality?
 84. What is the Laplace correction?
 85. What is "Parzen Windows"?
 86. How does Parzen Windows estimate a probability density?
 87. How does k-Nearest Neighbors estimate a probability density?
 88. How can a Maximum-likelihood estimator be used to estimate a probability density?
 89. What is the ML estimate of the parameters of a Gaussian distribution?
 90. What is the basic difference between a ML estimator and a Bayesian estimation of probability density?
 91. In which case will the ML estimator and the Bayesian estimation lead to the same results?

92. What are graphical models?
93. What are typical tasks solved with graphical models?
94. What is conditional independence?
95. How to derive a graphical model given a factorization of the joint probability?
96. How to derive a factorization of the joint probability given a graphical model?
97. What is a DAG?
98. What does MRF stand for?
99. What is a clique?
100. What is a maximum clique?
101. What is the general factorization for a Bayesian Network?
102. What is the general factorization for a MRF?
103. What is the relationship between the potential and the energy of a MRF?
104. What are unary (association) and binary / pairwise (interaction) energy terms (potentials)?
105. How does a typical MRF for image processing look like?
106. What does CRF stand for?
107. What is the basic difference between MRF and CRF?

108. What are the weights in an MLP?
109. What is the bias term in an MLP?
110. What are common activation / transfer functions in an MLP?
111. Which properties do transfer functions in an MLP usually have?
112. What is the update rule for the weights in an MLP?
113. What is the difference between online, batch, and offline learning?
114. What is momentum / inertia and weight decay?
115. What is meant by “receptive field” and “shared weights” in the context of ConvNets?
116. Which activation function is usually used for ConvNets?
117. What is the basic architecture / layout of a ConvNet?
118. What is dropout?
119. What is meant with “data augmentation”?
120. How does pre-training a ConvNet work?